Human Cellular Aging Project Tasks

THE IDEA:

An active area of bioinformatics R&D involves developing therapies to treat age-related diseases by reversing the biological clock of old cells through partial epigenetic reprogramming. As cells age, they experience a degradation of epigenetic marks, causing them to lose their youthful function and regenerative capacity. The following technical and therapeutic challenges to identifying transcription factors that can help change a cell to a more youthful state include:

1. **Partial Epigenetic Reprogramming:** Their main scientific hurdle is finding precise combinations of transcription factors (proteins that control gene activity) that can reset an old cell's epigenome to a youthful state. The goal is to restore a cell's original, healthy function without completely erasing its identity or reverting it into a pluripotent stem cell, which could otherwise cause tumors.

2. **The AI Discovery Engine:** Because the design space for testing combinations of transcription factors is overwhelmingly massive, biotech and bioinformatic companies often use proprietary AI systems that run in-silico experiments to predict how different factor combinations will change the phenotype of an old cell, exponentially speeding up real-world laboratory screening.

3. **LNP-mRNA Delivery Systems:** Some companies are designing lipid nanoparticle (LNP)-delivered RNA medicines. These temporarily express the youth-restoring transcription factors in target tissue without altering the patient's permanent DNA architecture.

THE TASK:

Use single-cell RNA sequencing (scRNA-seq) data to identify rejuvenation proteins using a computational pipeline that compares young and old cellular states, maps regulatory networks, and simulates genetic perturbations.

1. Define the Aging Transcriptomic Signature

The first step is to establish what "young" and "old" look like at the single-cell level.

- **scRNA-seq File Information:**
    - Go to the “Cell Aging” shared folder (https://drive.google.com/drive/folders/1_y2Jj-R9n8_xV_ifoIbqlWGgoFYBfVg-)and download the datasets in that folder.

    - The “metadata.json” file describes each “Sample_<#>.tar” file. Aftea for un-taring each .tar file you should see a folder with three files, a Sample_<#>_matrix.mtx file, a Sample_<#>_feature.tsv file, and a Sample_<#>_barcodes.tsv file.
    - You have been provided Python code that shows you how to read the .mtx and .tsv files and convert it to a .h5ad file and save it. Do that for all files.
    - The “metadata.json” file also indicates which Sample file is for a young or elderly person.
- **Perform a scRNA-seq Analysis of The .h5ad files**:
    - Save the converted .h5ad files in the “data” folder inside of your PWD.
    - Create the first part of your pipeline to read in each .h5ad file and conduct a scRNA-seq QC, normalization, …, etc analysis using Claude-Sonnet-5 as demonstrated to date.
    - Make Claude-Sonnet-5 save the result as a new and updated .h5ad file in a ./output folder inside of your PWD. <- I suggest giving each saved .h5ad file a new name slightly different from the input .h5ad file that is analyzed.
- **Build scBubbleTrees:**
    - Build a scBubbleTree for each analyzed .h5ad ./output file. Example scBubbleTree building code has been provided.
    - Use the scBubbleTrees to conduct the following steps.
- **Differential Expression (DE):** Use the scBubbleTrees to compare young and old cell populations within the same cell type to find genes that change significantly with age.
- **Age-Associated Markers:** Identify upregulated genes (associated with senescence, inflammation, or dysfunction) and downregulated genes (associated with metabolism or proteostasis).

2. Infer Gene Regulatory Networks (GRNs)

Since you want to find regulatory proteins (like transcription factors) rather than just structural genes, you need to map how genes control each other.

- **TF Activity Inference:** Use tools like **SCENIC** (Single-Cell Regulatory Network Inference and Clustering) to identify active transcription factors and their target genes (regulons).
- **Network Topology:** Pinpoint "hub" proteins that control large modules of age-associated genes.

3. Model Cellular Trajectories and Pseudotime

Aging is a continuous process rather than a binary switch.

- **Pseudotime Analysis:** Use tools like **Monocle3** or **Palantir** to line up cells along a continuous trajectory from young to old.
- **Reprogramming Trajectories:** If you have data from cells undergoing reprogramming (e.g., Yamanaka factor expression), map the trajectory to see which proteins drive the cells backward along the aging axis.

4. Conduct *In Silico* Perturbation Screening

Instead of testing thousands of proteins in a wet lab, use computational simulations to predict which proteins can reverse the aging signature.

- **Network Perturbation:** Use tools like **CellOracle** (https://morris-lab.github.io/CellOracle.documentation/) to simulate the over-expression or knockout of specific transcription factors. The algorithm predicts how the entire gene network will shift.
- **Deep Learning Models:** Use foundation models for single-cell biology (such as **scGPT** (https://github.com/bowang-lab/scgpt -and- https://www.nature.com/articles/s41592-024-02201-0) or **GEARS** (https://github.com/snap-stanford/GEARS -and- https://gearsblog.blogspot.com/2011/03/stopping-gears.html) to predict the transcriptomic outcome of perturbing specific genes.
- **Reversal Scoring:** Look for perturbations that shift the predicted expression profile of an old cell closest to the actual profile of a young cell.

5. Prioritize and Validate Candidates

Filter your computational hits down to a manageable list for laboratory validation.

- **Druggability & Feasibility:** Cross-reference your candidate proteins with databases (like DGIdb) to see if small molecules or gene therapies can target them.
- **Experimental Validation:** Validate the top protein candidates using CRISPR activation (CRISPRa), RNAi, or small molecules in old cell cultures, then measure rejuvenation using biological clocks (like epigenetic clocks or transcriptomic aging clocks).